Project Initialization and Planning Phase

Date	30 June 2026
Team ID	xxxxxx
Project Title	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal aims to transform agricultural production using AI-driven crop recommendations, boosting farm efficiency and yield accuracy. It addresses the lack of data-driven crop advisory tools for farmers by providing intelligent, soil-and-climate-aware guidance. Key features include a machine learning-based crop recommendation engine and real-time environmental suitability assessment.

Project Overview	
Objective	The primary objective is to develop an intelligent, data-driven crop recommendation system (OptiCrop) that uses soil parameters and environmental factors — including Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall — to recommend the most suitable crop for a given field, maximizing yield and resource efficiency.
Scope	The project encompasses data collection and preprocessing of soil and environmental datasets, ML model development for crop recommendation and suitability assessment, a web-based interface for farmers and researchers, and a policy analytics module for agri-planners.
Problem Statement	
Description	Farmers currently lack access to personalized, data-driven crop advisory tools. Decisions are made based on tradition or generic advice, often ignoring critical soil nutrient levels (N, P, K), pH, rainfall, and climatic factors — leading to poor crop selection, resource wastage, and reduced agricultural productivity.
Impact	Solving this problem will result in improved crop yields, reduced fertilizer waste, optimized resource usage, better income for farmers, and data-backed agricultural policy decisions for researchers and government planners.
Proposed Solution	
Approach	Employing supervised machine learning techniques to analyze soil and environmental parameters and recommend the best-fit crop. The system will also assess crop-soil-climate suitability for any chosen crop and provide fertilizer optimization guidance.

Key Features	- AI-powered crop recommendation engine using N, P, K, temperature, humidity, pH, rainfall. - Real-time soil-climate suitability assessment for any chosen crop. - Fertilizer and resource optimization module for cost reduction. - Web-based farmer interface with multilingual support. - Analytics dashboard for researchers and policymakers. - Historical yield tracking and sustainability scoring.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU / Intel Core i5+
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python ML frameworks	Flask, Streamlit
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Dataset	Source, size, format	Kaggle Crop Recommendation Dataset, 2200 rows, CSV UCI Agricultural Data, CSV